

Yifan Dong

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Profile

- I am a third-year Ph.D. student in Electrical and Computer Engineering with a strong academic background and hands-on experience in optimization and machine learning for power systems. I am skilled in programming languages such as Python and MATLAB. I am looking for an internship for Summer 2026.

Education

Purdue University, West Lafayette, IN

Aug 2023 – Present

Ph.D. in Electrical and Computer Engineering

- Research interest: Optimization and machine learning for power systems
- GPA: 3.84/4.0 [Transcript]
- Coursework: Computational Method in Optimization, Convex Optimization, Computational Method for Power System Analysis, Optimization for Deep Learning, Algorithm Design, Analysis and Implementation, etc.

North China Electric Power University, Beijing, China

Aug 2019 – June 2023

B.Eng. in Electrical Engineering and Automation

- GPA: 4.31/5.0 (Top 1%) [Transcript]
- Coursework: Power System Analysis, Power System Economy and Management, Automatic Control Theory, etc.

Publications

- Yifan Dong, Ge Chen, Junjie Qin, "**Federated Aggregation of Demand Flexibility**", *IEEE Transactions on Smart Grid*, 2026.
- Yifan Dong, Ge Chen, Junjie Qin, S. Sivaranjani, Xiaonan Lu, Dionysios Aliprantis, David Love, "**Real-Time Charging Control for Electric Roadways**", *IEEE Transactions on Smart Grid*, 2026.
- Yifan Dong, Junjie Qin, S. Sivaranjani, Xiaonan Lu, Dionysios Aliprantis, David Love, "**Real-Time Charging Control for Electric Roadways: Formulation and Causal Algorithms**", *IEEE Power & Energy Society General Meeting*, 2024.

Experiences

Research Assistant: ASPIRE Research Center

Aug 2023 – Present

- *Project 1: Privacy-Preserving Demand Flexibility Aggregation for Flexible Loads*
 - Designed a **distributed, privacy-preserving system** inspired by **federated learning**, enabling large-scale coordination among devices while keeping user data local and secure.
 - Re-architected a complex bilevel optimization model into a **scalable single-level learning workflow**, improving system reliability, maintainability, and end-to-end performance.
 - Developed a **decentralized, gradient-based algorithm** with backend logic for secure exchange of anonymized model parameters, reducing communication overhead and preserving data privacy.
 - Conducted extensive testing and performance evaluation, **demonstrating significant improvements in system efficiency and aggregate flexibility** over existing baselines.
 - Tech stack: Python, PyTorch, CvxpyLayer, NumPy
- *Project 2: Real-Time Charging Optimization System for Electric Roadways*
 - Designed and implemented a **real-time charging control system** for electric roadways, enabling safe, efficient wireless power transfer between roadway microgrids and electric vehicles under dynamic traffic conditions.
 - Re-architected the charging-control problem using **dynamic programming (DP)**, exposing **state-transition structure and constraints** that enabled a simplified and implementable solution strategy.
 - Developed a **causal, online scheduling algorithm** that exploits the DP-derived properties to support **lightweight, real-time decisions** without relying on future information.
 - Evaluated the system across diverse traffic scenarios, demonstrating **low suboptimality, reliable constraint satisfaction, and strong robustness to prediction errors**.
 - Tech stack: MATLAB, Python, SUMO

Teaching Assistant: Purdue University

Fall 2024, Spring 2026

- *ECE 31032: Power Systems Engineering*
 - Supported instruction on economic dispatch, transformers, power system modeling and power flow analysis.